L-T-785C
April 13, 1977
SUPERSEDING
Int. Fed. Spec. L-T-00785B (GSA-FSS)
October 12, 1970 and
Fed. Spec. L-T-785A
October 23, 1965
(see 6.3)

# FEDERAL SPECIFICATION

# TUBES, DRINKING, (PLASTIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rigid plastic drinking tubes (straws) and flexible-section tubes suitable for hot or cold liquids (see 6.1).

1.2 Classification. Plastic drinking tubes shall be of the following types, styles, and lengths, as specified (see 6.2):

Type I - Rigid.
  Style 1 - Slim.
    Length - 5-7/8 inches, plus or minus 1/8 inch.
  Style 3 - Super jumbo.
    Length - 7-7/8 inches, plus or minus 1/8 inch.

Type II - Flexible section.
  Length - 8-1/4 inches, plus or minus 1/4 inch.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

FSC 7350

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

Laws and Regulations:

21 CFR 121.2501 - Food Additives Resulting From Contact With Containers or Equipment and Food Additives Otherwise Affecting Food; Olefin Polymers.

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal Agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The material for the tubes shall be clean, sanitary, and free from toxic ingredients and shall conform to 21 CFR 121.2501.

3.2 Construction, type II. The type II flexible section tube shall be 8-1/4 ± 1/4 inches long with one arm 5-3/4 ± 1/8 inches and the other arm 2 ± 1/8 inches long. Between the arms shall be a corrugated section 1/2 inch, minimum, allowing the tube to be adjustable by bending to any angle through an arc of 130 degrees. The tube shall retain the angle to which it is bent.

3.3 Color. The color of the tubes shall be clear or white.

3.4 Dimensions.

3.4.1 Inside diameter. The inside diameter of the plastic drinking tubes shall be as shown below:

Inside Diameter of Tubes (inches)

| | Minimum | Nominal | Maximum |
|---|---|---|---|
| Type I | | | |
| Style 1 | 0.140 | 0.150 | 0.160 |
| Style 3 | 0.235 | 0.245 | 0.255 |
| Type II | 0.208 | 0.218 | 0.228 |



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3.4.2 Wall thickness. The wall thickness of the tubes shall be as follows:

| | |
|---|---|
| Type I | |
| Style I | 0.005 ± 0.0005 inch |
| Style 3 | 0.007 ± 0.0005 inch |
| Type II | 0.007 ± 0.0005 inch |

3.5 Rigidity. The tube shall not collapse or distort when tested in accordance with 4.4.1.

3.6 Softening, type II tube. The tube shall be useable for hot liquids up to 150°F., without distortion or impairment due to softening of the plastic, when tested as specified in 4.4.2.

3.7 Odor and taste. The tube shall be odor free and shall not impart any extraneous taste to the liquid when tested in accordance with 4.4.3.

3.8 Wrap. The tube shall be individually wrapped or unwrapped, as specified (see 6.2). When wrapped, each tube shall be sealed in a wrap for protection against contamination in handling. Commercial colors and markings are acceptable on the wrap.

3.9 Workmanship. The tubes shall be clean, well made, and free from defects detracting from appearance and impairing serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling for inspection and acceptance shall be made in accordance with MIL-STD-105.

4.2.1 Sampling for inspection. Samples shall be selected in accordance with level S-3 (see 4.3.1).

4.2.2 Sampling for tests. Samples for tests shall be in accordance with level S-2. The Acceptable Quality Level (AQL) shall be 2.5 percent defective.

4.3 Inspection.

4.3.1 End item. Tubes selected in 4.2.1 shall be examined for defects listed in table I. The acceptable quality level (AQL) shall be 4.0 percent defective.

TABLE I. Classification of defects, end item

| Examine | Defects |
|---|---|
| Wrapping (as applicable) | Omitted; not sealed. |
| Appearance | Color not as specified.<br>Dirt, plastic dust, foreign material. |
| Dimensions | Incorrect length.<br>Length of type II arms not as specified.<br>Incorrect inside diameter.<br>Incorrect wall thickness. |
| Construction | Type II flexible section not adjustable as specified. |
| Workmanship | Rough surface and ends.<br>Cross section not reasonably circular.<br>Length not reasonably straight. |



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4.3.2 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and marking requirements comply with section 5. Defects shall be scored in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of a closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE II. Examination of packing

| Examine | Defects |
|---|---|
| Markings (exterior and interior) | Omitted; illegible; improper size, location, sequence, or method of application. |
| Materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container. |

4.4 Tests.

4.4.1 Rigidity. Immerse one end of the tube 2-1/2 to 3 inches in water having a temperature of 40° ± 2°F. for 5 minutes. The longer arm of the type II tube shall be immersed. Approximately three ounces of water shall then be siphoned through the tube to determine compliance with 3.5 (see 4.4.3).

4.4.2 Softening. Place the type II tube in an uncovered container of water at 150° + 0°F. - 2°F., for 5 minutes. Tube shall comply with 3.6.

4.4.3 Odor and taste. The tube tested in 4.4.1 shall be checked for odor and the water tested for conformance to 3.7.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or commercial as specified (See 6.2).

5.1.1 Level A. Five hundred type I, style 1 or four hundred type I, style 3 or two hundred type II tubes of like description shall be packaged in a close fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. The box shall be closed in accordance with the applicable box specification or appendix thereto.

5.1.2 Commercial. Tubes shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, or commercial as specified (see 6.2).

5.2.1 Level A. Tubes of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, grade V2s of PPP-B-636. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636. The weight of contents of each fiberboard container shall not exceed 65 pounds.

5.2.2 Commercial. The tubes shall be packed to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with Uniform Freight Classification or National Motor Freight Classification.



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5.3 Marking. In addition to any special markings required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Tubes are intended for drinking hot or cold liquids. The type II flexible section tube is intended to facilitate drinking by bed patients.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of tube (see 1.2).
(c) Whether tube shall be wrapped or unwrapped (see 3.8).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) Marking required (see 5.3).

6.3 Supersession data. The classification of the tubes between this revision and the previous edition has been changed. The types have been redesignated to styles under one type (type I - Rigid). Wrapping of the tubes is now given as an option in 6.2. The lengths as stated in L-T-00785B no longer apply. A flexible section tube as type II has been added. Comparison of classification between these documents is as follows:

| L-T-00785B | L-T-785C |
|---|---|
| Type I - Slim, unwrapped | Type I - Rigid - Style 1 - Slim |
| Type II - Slim, individually wrapped | Type I - Rigid - Style I - Slim |
| Type II - Jumbo, unwrapped | Deleted |
| Type IV - Jumbo, individually wrapped | Deleted |
| Length 1 - 6-1/4 inches, plus or minus 1/8 inch | Length - 5-7/8 inches, plus or minus 1/8 inch |
| Length 2 - 8-1/4 inches, plus or minus 1/8 inch | Deleted |
| Length 3 - 10-1/4 inches, plus or minus 1/8 inch | Deleted |
| Type V - Super jumbo, individually wrapped | Type I - Rigid<br>Style 3 - Super jumbo |
| Length 2 - 8-1/4 inches, plus or minus 1/8 inch | Length - 7-7/8 inches, plus or minus 1/8 inch |
| Length 3 - 10-1/4 inches, plus or minus 1/8 inch | Deleted |
| Type IV - Giant, Individually wrapped | Deleted |
| Length 2 - 8-1/4 inches, plus or minus 1/8 inch | Deleted |
| | Added<br>Type II - Flexible section<br>Length - 8-1/4 inches, plus or minus 1/4 inch |



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6.4 Inclusion. The requirements of Military Specification MIL-T-36610 have been included in this document.

MILITARY INTEREST:

Custodian:

Army - GL

Review activities:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

GSA-FSS
VA-DMS

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